

1. Administrative & Study Identification

Full Study Title: Korean Post-marketing Surveillance Ngenla® Pre-filled Pen Injection for the Treatment of Pediatric Patients With Growth Disturbance Due to Insufficient Secretion of Growth Hormone **Short Title/Acronym:** Ngenla® PMS Korea **Protocol Number:** Pfizer Internal PMS ID **PostingID:** 380284580193534615 **NCT Number:** NCT05509894 **Sponsor Name / Company:** Pfizer **Investigational Product:** Ngenla® Pre-filled Pen Injection (somatrogen) **Trade Names:** Ngenla **ATC Code:** H01AC08 **Phase of Development:** Not Applicable **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) **Medical Monitor:** Pfizer CT.gov Call Center

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and effectiveness of Ngenla® Pre-filled Pen Injection for the treatment of pediatric patients aged 3 years and older with growth disorders due to insufficient secretion of growth hormone over a 6-year period in a routine real-world practice setting in Korea. **Secondary Objectives:** To monitor adverse events, treatment compliance, and evaluate long-term outcomes such as annual height velocity and height standard deviation score in the post-marketing environment.

2.2 Background & Rationale This non-interventional study is conducted as a Post-Marketing Surveillance (PMS) study to fulfill the additional pharmacovigilance activity of the Risk Management Plan (RMP) for the Ngenla® pre-filled pen. This surveillance is required by the Ministry of Food and Drug Safety (MFDS) in accordance with local regulations in Korea to continuously track the safety profile of the once-weekly subcutaneous injection in the pediatric population.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Non-interventional, Post-marketing surveillance) **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Targeted Enrollment (\$N\$): 565 evaluable pediatric patients **Number of Sites:** Routine pediatric endocrinology clinical practice sites across Korea **Estimated Study Start:** 20-Jun-2024 **Estimated Primary Completion:** Jun-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Children and adolescents from 3 years of age with growth disturbance due to insufficient secretion of growth hormone who have received treatment with somatropin or have been determined to start treatment with somatropin according to the approved indications of the medicinal product; 2. Evidence of a personally signed and dated informed consent document indicating that the patient or their parent(s)/legal guardian, if applicable, have been informed of all pertinent aspects of the study.

4.2 Exclusion Criteria 1. Patients concurrently participating in other studies involving therapeutic interventions and/or investigational products; 2. Patients who have contraindications to somatropin as specified in the approved LPD. 3. Patients with hypersensitivity or case history to somatropin or to any of the excipients in the product.

5. Intervention & Treatment Plan

5.1 Investigational Regimen **Drug Name:** somatropin **Dosage/Frequency:** 0.66 mg/kg body weight administered once weekly. **Route of Administration:** Subcutaneous injection. **Duration of Treatment:** Long-term treatment with up to 6 years of observational follow-up.

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant medications for pediatric care and comorbidities as prescribed by the HCP in routine clinical practice. **Prohibited:** Concurrent use of other recombinant human growth hormone (rhGH) therapies or treatments contraindicated per the local Ngenla® product label.

6. Study Timeline & Visit Schedule

(Note: As this is an observational real-world study, study visits correspond to standard routine clinical practice).

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Clinical Diagnosis, Prescription of Ngenla®, Informed Consent
Baseline	Day 0	First Dose, Medical History, Baseline Auxological Assessment
Interim		Routine Care Periodic Onsite Visits: Safety Assessment, Height/Weight Measurements, Compliance Check
End of Study	Up to Year 6	Final Auxological Assessment, Overall Safety Review

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. Incidence of adverse events (AEs)

7.2 Secondary Outcomes 1. Change from baseline in Annualized Height Velocity in cm/year 2. Change from baseline in Height Standard Deviation Score 3. Change from baseline in Bone Maturation (BM)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard pharmacovigilance data collection via medical records during routine clinical visits. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness to the sponsor and the Ministry of Food and Drug Safety (MFDS). **Data Monitoring Committee (DMC):** Supervised internally by the Pfizer Pharmacovigilance and Risk Management team.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all enrolled patients who received at least one dose of Ngenla®). **Sample Size** **Justification:** Target sample of $N=565$ established and agreed upon with the MFDS to provide an adequate representation for post-marketing safety surveillance. **Handling of Missing Data:** Analyzed as observed cases; no imputation methods (like LOCF or MI) are typically utilized for missing real-world safety data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Pharmacoepidemiology Practices (GPP), local MFDS post-marketing regulations, and the Declaration of Helsinki. **Monitoring Plan:** Routine data quality checks and periodic source data verification based on the post-marketing surveillance framework. **Audit Trail:** Secure electronic data capture (EDC) system featuring comprehensive audit trailing and data clarification mechanisms.

11. Study Identifiers & Governance

Primary ID: {{NCT05509894}} **Registry Number:** {{NCT05509894}} **Sponsor/Lead Organization:** {{Pfizer}} **Study Title:** {{Korean Post-marketing Surveillance Ngenla® Pre-filled Pen Injection}} **Official Regulatory Title:** {{Post Marketing Surveillance (PMS) Study for Ngenla Prefilled Pen in Pediatric Patients Who Have Endogenous Growth Failure Due to an Inadequate Secretion of Endogenous Growth Hormone in Korea}}

12. Clinical Framework (Condition Specific)

Conditions / Disease Areas: {{Pediatric Growth Hormone Deficiency}} **Preferred UMLS Name:** {{Pediatric growth hormone deficiency}} **Diagnosis Threshold:** {{Insufficient secretion of endogenous growth hormone}} **Medical Methodology:** {{Long-acting Growth hormone replacement therapy}} **Optimization Target:** {{Age-appropriate height velocity and growth normalization}}

13. Study Procedures & Methodology

Management Pathway: {{Standard pediatric endocrinology clinical pathway for growth disorders}} **Education Protocol (HCP):** {{Proper dosing calculation and administration training for once-weekly Ngenla pre-filled pen}} **Education Protocol (Patient):** {{Caregiver and patient training on subcutaneous self-injection technique, site rotation, and compliance}} **Quality Audit Mechanism:** {{Routine pharmacovigilance monitoring and RMP compliance tracking}}

14. Observation & Follow-Up Schedule

Total Duration: {{Up to 6 years (72 months)}} **Onsite Visit Cadence:** {{As per routine clinical practice schedule}} **Remote Monitoring Frequency:** {{N/A - Observational}} **Checklist Review Interval:** {{Periodic review of auxological measurements and AE occurrence}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				

Appendix: System Metadata & Database Mappings

To ensure full database alignment, the following background system mapping properties apply to this protocol record: * **PostingID:** 380284580193534615 * **Phase ID:** Not Applicable * **Status Name:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) * **Criteria Type:** PHASE_REQUIREMENT * **Criteria Text:** Trial must be in Not Applicable * **Drug Preferred Name:** somatrogen * **Drug Semantic Type:** [Empty/None] * **ATC Codes:** H01AC08 * **RxNorm Code:** [Empty/None] * **EMA URL:** https://www.ema.europa.eu/en/documents/product-information/ngenla-epar-product-information_en.pdf * **Source Level:** 5 * **Ancillary System Mappings (Ontology Cross-References):** * *Name:* B-cell Non Hodgkin Lymphoma * *Preferred*

Name: Indolent B-Cell Non-Hodgkin Lymphoma * *Semantic Type:* Neoplastic Process * *Definition:* A B-cell non-Hodgkin lymphoma with an indolent clinical course. Representative examples include grade 1 and grade 2 follicular lymphoma, marginal zone lymphoma, and lymphoplasmacytic lymphoma. * *MeSH Code:* D008182 * *SNOMED ID:* 92000000 * *SNOMED Hierarchy:* 128361009|Non-Hodgkin lymphoma|Non-Hodgkin lymphoma (disorder)